Workspace

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Data Cleaning

The goals of this section:

- punctuation
- filler words (ex. oh, ah, er)
- wrong glossing (as in proper nouns labeled as nouns or NAs)
- mixed-language labeling (also names)

```
all_data <- full_join(herring1_cgwords, herring2_cgwords)
```

```
Joining with `by = join_by(word_id, utterance_id, location, surface, auto, fix,
eng, com, speaker, langid, filename, clause, clauseno)`
```

```
#This step removed columns that are composed of entirely NA values. it then summarizes the nu
data <- herring1_cgwords |>
  select(-fix,-eng, -com, -clause, -clauseno) |>
  group_by(langid) |>
  summarize(n())

#The prior step shows that punctuation is coded as 999 under language, which makes it unusefu

#Remove punctuation and remove the redundant auto gloss repetition of the word.
data_clean <- all_data|>
  select(-fix,-eng, -com, -clause, -clauseno) |>
  filter(langid != "",langid !="999" ) |>
  mutate(auto = if_else(str_detect(auto,"."), str_extract(auto,"(?<=\\.).*"), auto),
         pos = if_else(str_detect(auto,"."), str_extract(auto,"^[^.]+"), auto)
```

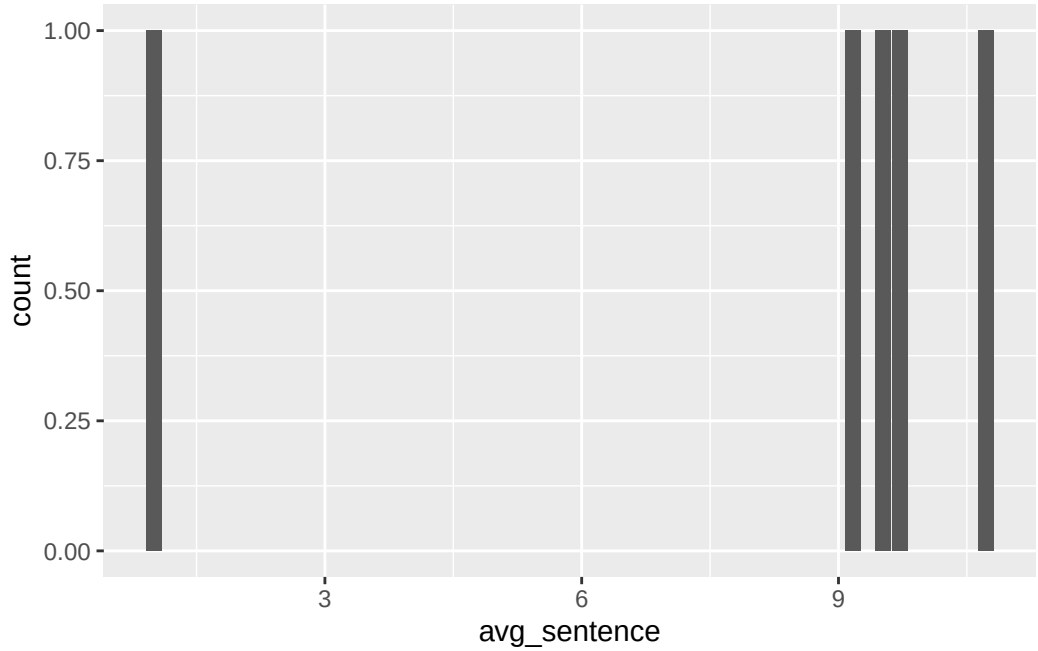
```
        )

#extract major POS, separate inflections
data_clean <- data_clean |>
  mutate(
    langid = str_replace(langid,"eng&spa", "eng"), # might change to remove
    pos = na_if(pos, ""),
    pos = replace_na(pos, "N"),
    gender = if_else(str_detect(auto,"\\.(F|M)\\."), str_extract(auto,"(F|M)"), "neuter"),
    number = if_else(str_detect(auto,"\\.(SG|\\dS|P)"), str_extract(auto,"(SG|\\dS|\\dP)"),

# Joining for superdata on conversations









# test_str <- c("PRON.SUB.F.3S","V.INFIN")
#
# if_else(str_detect(test_str,"\\.(F|M)\\."), str_extract(test_str,"\\.(F|M)"), "neuter")
# str_extract(test_str ,"\\.(F|M)")
```

Summary Statistics

Goals:

- determine where switches happen (before and after)
- Spanish use per speaker
- Frequency of code-switch per speaker and part of speech

```
#utterance word length by speaker
data_clean |>
    add_count(speaker, utterance_id) |>
    group_by(speaker) |>
    summarise(avg_sentence = mean(n)) |> # keep until here for avg utternace length
  ggplot(aes(x=avg_sentence)) + # prob remove not that useful
  geom_bar() #remove OSE
```




```
#Length of utterance
# ggplot(data_clean, aes(x=utterance_id)) +
#   geom_bar()

#spanish frequency by speaker
data_clean |>
  group_by(speaker) |>
  filter(langid == "spa") |>
  summarize(n())
```

```
# A tibble: 4 x 2
  speaker `n()`
  <chr>   <int>
1 CHL       161
2 LAU       130
3 MIG      3542
4 TOM      1693
```

```
#POS 1 before switch
boundaries <- data_clean |>
  group_by(speaker, utterance_id) |>
  mutate(
```

```
    next_lang = lead(langid), #find the next entry for lang
    next_pos = lead(pos) #next entry for POS
  ) |>
  ungroup() |>
  filter(!is.na(next_lang)) |> #Utterance end filter (speaker change)
  mutate(switch = as.integer(langid != next_lang)) #binary codeswitch

count(boundaries, pos, sort =TRUE)
```

```
# A tibble: 37 x 2
   pos       n
   <chr> <int>
 1 V      2081
 2 PRON   1836
 3 CONJ   1315
 4 N      1065
 5 PREP   1001
 6 ADV     871
 7 ADJ     662
 8 DET     658
 9 NUM     104
10 IM       90
# i 27 more rows
```

```
pos_table <- boundaries |>
  group_by(pos) |>
  summarise(
    boundaries = n(), #how many times that POS appears
    switches   = sum(switch), #switch occurred after POS
    rate       = switches / boundaries,
    .groups = "drop"
  )

overall_rate <- mean(boundaries$switch)


Spk_rate_table <- boundaries |>
  group_by(speaker) |>
  summarise(
    boundaries = n(),
    switches   = sum(switch),
```

```
    rate       = switches / boundaries,
    .groups = "drop"
  )
```

Base Comparison

This section works to form the baseline comparison between experimental and corpus data. It investigates the ability of determiners to predict switches and how cognates and non-cognates further inform this.

Goals:

-

Possible statistical test:

markov chains?

What are the most common code-switched words (POS)?

Summary statistic + visualization +

```
# mlm <- glmer(switch ~ pos + location + (1|speaker) , data = boundaries, family = binomial(l
#
# summary(mlm)
```

Therefore making a predicition three or another clustering method in machine learning?

When will a switch happen?

The Special Case of Maria

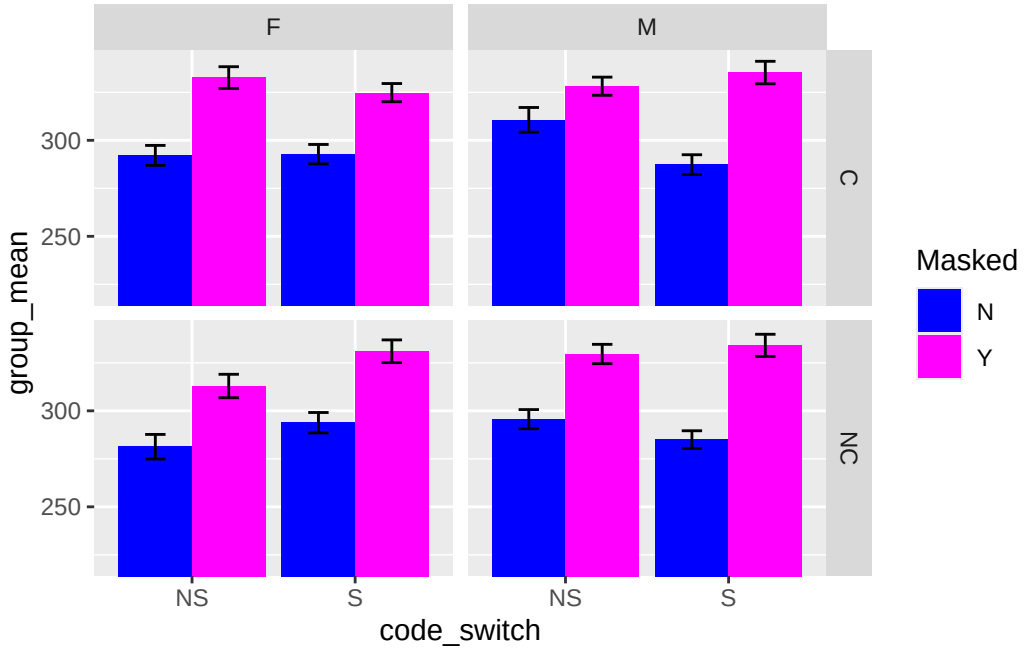
Something along the lines of

what part of speech is most relevant to code-switching prediction

Experimental Eye-Tracking Data

"The current study aimed to assess whether bilinguals similarly use visual cues in the parafovea, such as orthographic form and language status, to anticipate and therefore reduce costs related to the processing of an intra-sentential code-switch"

```r
test <- et_data |>
  group_by(cognate_status, code_switch, gender, Masked) |>
  filter(Article == 1|Target == 1, IA_SKIP == 0) |>
  summarize(
    group_mean = mean(as.numeric(IA_FIRST_RUN_DWELL_TIME)),
    SD = sd(as.numeric(IA_FIRST_RUN_DWELL_TIME)),
    n = n(),
    SE =SD/sqrt(n),
    .groups = "drop")

ggplot(test, aes(x = code_switch, y = group_mean, fill = Masked)) +
  geom_bar(stat = "identity", position = position_dodge(0.9)) +
             geom_errorbar(
               aes(ymin = group_mean - SE, ymax = group_mean + SE),
               position = position_dodge(0.9),
               width = 0.25                    # Width of the horizontal error bar caps
             ) +
  facet_grid(cognate_status ~ gender) +
  scale_fill_manual( values = c("Y" = "magenta", "N" = "blue")) +
  coord_cartesian(ylim = c(220,NA))
```



Cognate-Non Cognate target

Placing factors such as code-switching behavior, verbal fluency, cognitive control to code-switch anticipation